



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

OBJECT ORIENTED PROGRAMMING

LAB Assignment 3

SUBMITTED BY:

Name: Mridul Sharma

Roll no: 1024150204

Batch: 2023

Question1: Write a program to implement

(a) pointer to an object

(b) this pointer.

Practice both. (dot operator) and -> (arrow operator).

INPUT	OUTPUT
<pre>#include <iostream> using namespace std; class Sample { int x; public: void set (int x) { this->x = x; } void show () { cout<< "Value of x = " << x << endl; } }; int main() { Sample obj; Sample *ptr; ptr = &obj; obj.set(40); obj.show(); ptr->set(50); ptr->show(); return 0; }</pre>	<pre>Value of x = 40 Value of x = 50 === Code Execution Successful ===</pre>

Question2: Write a program to swap private values of two classes using a friend function.

INPUT	OUTPUT
<pre>#include <iostream> using namespace std; class B; class A { int x; public: A(int a) { x = a; } friend void swap(A&, B&); void show() { cout << "A = " << x << endl; } }; class B { int y; public: B(int b) { y = b; } friend void swap(A&, B&); }</pre>	<pre>A = 42 B = 23 === Code Execution Successful ===</pre>

<pre> void show() { cout << "B = " << y << endl; } }; void swap(A &a, B &b) { int temp = a.x; a.x = b.y; b.y = temp; } int main() { A a(23); B b(42); swap(a, b); a.show(); b.show(); return 0; } </pre>	
--	--

Question3: Write a program to add data objects of two different classes using friend functions.

INPUT <pre> #include <iostream> using namespace std; class B; class A { int x; public: A(int a) { x = a; } friend int add(A, B); }; class B { int y; public: B(int b) { y = b; } friend int add(A, B); }; int add(A a, B b) { return a.x + b.y; } int main() { A a(78); B b(22); cout << "Sum = " << add(a, b) << endl; return 0; } </pre>	OUTPUT <pre> Sum = 100 === Code Execution Successful === </pre>
--	--

Question4: Write a program to demonstrate the working of friend class.

INPUT <pre>#include <iostream> using namespace std; class A { int x; public: A() { x = 50; } friend class B; }; class B { public: void show(A a) { cout << "Value of x = " << a.x << endl; } }; int main() { A a; B b; b.show(a); return 0; }</pre>	OUTPUT <pre>Value of x = 50 === Code Execution Successful ===</pre>
--	---

Question5: Write a program using Array of Objects to display area of multiple rectangles.

INPUT <pre>#include <iostream> using namespace std; class Rectangle { int length, breadth; public: void set() { cin >> length >> breadth; } void area() { cout << "Area = " << length * breadth << endl; } }; int main() { Rectangle r[3]; for(int i = 0; i < 3; i++) { cout << "Enter length and breadth: "; r[i].set(); } for(int i = 0; i < 3; i++) { r[i].area(); } }</pre>	OUTPUT <pre>Enter length and breadth: 10 20 Enter length and breadth: 12 15 Enter length and breadth: 96 45 Area = 200 Area = 180 Area = 4320 === Code Execution Successful ===</pre>
--	---

<pre> } return 0; } </pre>	
------------------------------------	--

Question6: Write a program to define function cube() as inline for calculating cube of a number.

INPUT <pre> #include <iostream> using namespace std; inline int cube(int x) { return x * x * x; } int main() { int n; cout << "Enter number: "; cin >> n; cout << "Cube = " << cube(n) << endl; return 0; } </pre>	OUTPUT <pre> Enter number: 15 Cube = 3375 === Code Execution Successful === </pre>
--	---

Question7: Write a program to pass an object as an argument and return the object from a function.

a) Use pass-by-value

INPUT <pre> #include <iostream> using namespace std; class Number { public: int x; }; Number modify(Number n) { n.x += 10; return n; } int main() { Number n1, n2; n1.x = 5; n2 = modify(n1); cout << "Original: " << n1.x << endl; cout << "Modified: " << n2.x << endl; return 0; } </pre>	OUTPUT <pre> Original: 5 Modified: 15 === Code Execution Successful === </pre>
--	---

b) Use pass-by-address

INPUT

```
#include <iostream>
using namespace std;
class Number {
public:
    int x;
};
Number modify(Number *n) {
    n->x += 10;
    return *n;
}
int main() {
    Number n1, n2;
    n1.x = 5;
    n2 = modify(&n1);
    cout << "Modified: " << n2.x << endl;
    return 0;
}
```

OUTPUT

Modified: 15

=== Code Execution Successful ===